Technical report of AI versus unemployment project

Project idea

The most challenging aspect of this assignment was defining the requirements—both for the course as a whole and for the mini-project itself. The mini-project's assignment brief was too open-ended, failing to clearly highlight the learning objectives. These objectives could be categorized into technical implementation (EDA, data processing, analysis, etc.) and the planning and phasing of the project (defining the topic, requirements, target audience, communication, collaboration, etc.)—with the lectures and exercises focusing primarily on the former. Selecting the level of abstraction for the topic also proved difficult, as it was hard to assess its relationship to the assignment brief and the grading criteria.

Despite these difficulties, we managed to find a suitable topic for the mini-project. We set out to consider, in general terms, what we believe is wrong with the world—and with Finland in particular. After a couple of rounds of iteration, we ended up examining unemployment and how artificial intelligence—a topic on everyone's lips—might affect jobs in the future. This choice of topic allowed us to clearly define the subject (unemployment), the target audience (government decision-makers), and the format in which the project information should be presented (a report). These elements, in turn, established a natural framework for the methods employed and the acquisition of data, which are described in greater detail in the following sections. A potential weakness in the choice of topic is the heavy emotional charge, as almost everyone has direct or indirect experience with unemployment. This creates challenges when presenting the study's findings to decision-makers, particularly if the model's projections are distinctly negative.

Data collection and preprocessing

Data was collected from two sources: Statistics Finland and Eurostat. The datasets that ended up being used in the study were: Statistics Finland occupation specific unemployment data (categorized by ISCO):
(https://pxweb2.stat.fi/PxWeb/pxweb/en/StatFin/StatFin__tyonv/12ti.px/),

Statistics Finland industry specific AI adoption data (categorized by NACE):
(https://pxweb2.stat.fi/PxWeb/pxweb/en/StatFin/StatFin__icte/14yc.px/)

and Eurostat employment data (used to recategorize AI adoption data from NACE to ISCO):
(https://ec.europa.eu/eurostat/databrowser/view/lfsa_eisn2/default/table?lang=en).

One problem with the data is that the AI adoption data uses NACE and the unemployment data uses ISCO, which means that it is not possible to directly compare the datasets.

In order to compare the datasets, an approximate mapping from NACE to ISCO was performed for the AI adoption data. The mapping used Eurostat employment data, which contains employed persons by both NACE Rev. 2 industry and ISCO-08 occupation. For each ISCO group o and NACE industry i, an employment weight was calculated as $w(o,i) = E(o,i) / \Sigma E(o,j)$, where $E(o,i)$ is the average number of employed persons in occupation o

and industry i during 2021–2025. The estimated AI adoption value for each ISCO group was then calculated as $AI(o,t) = \Sigma_i\, w(o,i) \times AI(i,t)$, meaning that industries employing a larger share of a given occupation had a larger effect on that occupation's estimated AI adoption value.

Some aggregation was required because the industry categories in the two datasets were not identical. NACE sections C, D and E were combined to match the C–E category, while F, H, I, J and M could be matched directly. The AI data separates NACE section G into industries 45–46 and 47, so these AI adoption values were averaged to match the single G category in the Eurostat data. The AI category L, N, S951 was approximated using employment in L and N because industry 951 was not available separately. OC0 and OC6 were excluded from the final analysis.

Visualisations

Five visualisations were selected to investigate the relationship between the adoption of artificial intelligence and unemployment, moving from an overview of general trends to occupational comparisons, statistical predictions and comparison of historical unemployment data to more recent data.

The first graph compares overall unemployment with the percentage of enterprises using AI over time. This visualisation was chosen to establish the current situation and identify whether the two variables appear to follow similar or opposing trends. Displaying both measures on the same graph makes it easier to observe changes over time and identify possible relationships. Separate y-axes are used because the variables have different scales. The graph is intended to identify potential associations rather than demonstrate that AI adoption causes changes in unemployment.

The second visualisation examines the relationship between AI adoption and unemployment across different occupational groups. This approach allows comparisons between groups and helps identify whether certain occupations experience larger increases or decreases in unemployment alongside changes in AI adoption. Indexing unemployment to a common base year makes relative changes easier to compare between groups of different sizes. Additionally, examining changes between 2023 and 2025 provides a more focused comparison of recent developments. This visualisation helps identify patterns that may be hidden when looking only at overall figures.

For the third visualization, an Ordinary Least Squares (OLS) regression model was used to examine the relationship between AI adoption and unemployment while accounting for occupational groups and years. The model generates predicted unemployment values, which can be compared with observed values to evaluate how well the model represents the data. This provides a statistical complement to the visual analysis and helps assess whether AI adoption contributes explanatory information beyond differences between occupations and years.

The fourth visualisation examines seasonal patterns in monthly unemployment for each occupational group. A centred annual moving average was calculated using a 2×12-month moving average filter to estimate the underlying unemployment trend. Percentage deviations

from this trend were then calculated and smoothed using a centred three-month moving average to reduce short-term noise. Plotting these deviations makes recurring seasonal patterns and unusual disturbances easier to identify. Separate panels were used to compare patterns before and during the generative AI adoption period, starting in 2023, while also showing the disruption associated with the COVID period. The deviations contain both seasonal and irregular variation, so they are not a pure measure of seasonality. This visualisation helps assess whether recent fluctuations differ visibly from historical patterns.

The fifth visualisation examines changes in the unemployment floor, defined here as the lowest monthly count of registered unemployed jobseekers within each complete calendar year. Each occupational group's annual minimum was indexed to its 2019 value, which was set to 100, allowing proportional changes to be compared across groups of different sizes. All groups were displayed on a single graph to show whether their minimum unemployment levels move together or diverge over time. This visualisation complements the seasonal analysis by highlighting longer-term changes that may be less visible in deviations from a moving average. Both visualisations use shaded regions to distinguish the COVID period from the generative AI adoption period.

The methods provide complementary perspectives: the first establishes the context, the second explores differences between occupational groups, the third evaluates the relationship using a statistical model and generates predictions, the fourth examines seasonal patterns and unusual fluctuations, and the fifth tracks changes in annual minimum unemployment levels. The findings must be interpreted cautiously. The analyses describe associations and historical patterns rather than establish causal effects, and enterprise AI adoption may not accurately represent exposure within individual occupations. Stable seasonal patterns do not rule out gradual effects, while rising annual minimum unemployment levels cannot identify their cause. Therefore, the results should be treated as exploratory evidence rather than definitive proof of AI-related unemployment.

Learnings

We learned that choosing the right visualisation is important for understanding the relationship between AI adoption and unemployment. We initially started with linear regression to examine the relationship between the two variables. However, we realised that a simple regression did not sufficiently capture the time aspect of the data when making predictions.

We also went through several iterations of the occupational group visualisations. Our goal was to find a way to compare groups with different unemployment levels while making changes over time easier to understand. We eventually settled on a scatter plot that displays the different occupational groups and their relationship between AI adoption and unemployment. This made it easier to see potential patterns and compare changes between groups.

Future Steps and Further Research

A few limitations should be addressed to improve the reliability of the results. More data from other countries than Finland covering a longer period would help determine whether the

observed patterns represent long-term trends or temporary economic fluctuations. Additionally, unemployment is influenced by many factors beyond AI adoption, including economic growth, inflation, interest rates, demographic changes and developments within individual industries. Including these variables could provide a more complete understanding of what is affecting unemployment. The relationship between industry-level AI adoption and occupational unemployment should also be examined more carefully, as enterprise adoption rates do not necessarily reflect how AI is used in specific jobs.

Future research should investigate whether occupations with higher AI exposure experience different employment outcomes over time. It would also be valuable to distinguish between job losses, the creation of new positions and changes in job tasks caused by AI. Using more detailed occupational data and stronger statistical methods could help separate the potential effects of AI from other economic influences. This would provide a clearer understanding of which occupational groups are most vulnerable to AI-related changes in employment.

Reflection

The concept behind the task was good, as the intention was to simulate a real-world situation in a small-group format. The size of the group was just right, because otherwise someone would have ended up with too little work. The canvas designed for tracking the project's progress was initially a bit cryptic, as I found it difficult to fit my ideas into it. The elevator pitch was also spot on when considering, for example, raising funding in the real world.

The clear division in the course structure—between technical implementation and project planning, with an emphasis on the former—raised questions about the purpose of the course: if it is intended for those who already possess strong programming skills, couldn't they have simply learned the required technical components on their own? This is because innovation and collaboration were not actively fostered; instead, they were viewed as innate traits that cannot be learned—unlike coding.

The freedom to choose one’s own topic did not foster innovation or collaboration either. Instead, it merely led to increased frustration and confusion, causing the actual subject matter to become obscured. Had the assignment been more constrained—for instance, by offering a choice between a few specific options—the aforementioned project planning and process stages would have been more prominent. This would have established a clearer, more logical connection between the use of specific techniques and the various stages of the project lifecycle. Is it truly "creativity" to invent both a problem and its solution, or is it more creative to devise a solution for a problem defined by someone else, drawing from the nearly limitless array of tools and approaches available in data science?

A more constrained approach would have also improved the peer-review experience; since everyone would have been grappling with a similar type of problem, participants could have learned from one another’s solutions. As it stood, the final presentation session felt more like a casual film club meeting focused on short films. One could also view it this way: it would be more pleasant to have more than just 3 minutes to present the results of one’s own work.

In this instance, the project outcome may have been forced too rigidly into a specific mold—one designed to deliver a "wow" effect. This, albeit unintentionally, imposes quite radical constraints on how the work is actually executed. We experienced this firsthand: the credibility of information presented to government decision-makers (in our case, a model) is undermined if it is forced into the format of an application, a webpage, or—even worse—a blog post.

In a professional context, one would generally expect to be given a specific task, rather than being handed a stack of cash and told, "Go do something."

This borders on course feedback, but it relates to understanding our own learning task and approach to it. We do not deny that something was learned from this assignment. Perhaps an increase in critical thinking—directed both toward oneself and toward others—can be considered a key takeaway from this exercise.